

Distribution of Gal alpha 1----3Gal beta 1----4GlcNAc residues on secreted mammalian glycoproteins (thyroglobulin, fibrinogen, and immunoglobulin G...

The study of the expression of Gal alpha 1----3Gal beta 1----4GlcNAc residues on mammalian glycoconjugates is of particular interest since as many as 1% of circulating IgG antibodies in man (the natural anti-Gal antibody) interact specifically with this carbohydrate residue. In recent studies, we have found that Gal alpha 1----3Gal beta 1----4GlcNAc residues are abundant on red cells and nucleated cells of nonprimate mammals, prosimians, and New World monkeys, but their expression is diminished in Old World monkeys, apes, and humans. In the present work, we have analyzed the expression of these residues on secreted mammalian glycoproteins. For this purpose, we have developed a radioimmunoassay (RIA) which enables the quantification of Gal alpha 1----3Gal beta 1----4GlcNAc residues on the secreted glycoproteins. Purified biotinylated anti-Gal was used as the antibody in the RIA, and bovine thyroglobulin enriched for Gal alpha 1----3Gal beta 1----4GlcNAc residues served as a solid-phase antigen. In this study, it is reported for the first time that the evolutionary pattern of Gal alpha 1----3Gal beta 1----4GlcNAc residue distribution in in vivo secreted glycoproteins is similar to that observed in membranes of cell lines and of red cells. Thyroglobulin, fibrinogen, or IgG molecules from nonprimate mammals and from New World monkeys express varying amounts of Gal alpha 1----3Gal beta 1----4GlcNAc residues ranging between 0.01 and 11 residues per molecule, whereas no such residues are present on any of these glycoproteins of human or Old World monkey origin.(ABSTRACT TRUNCATED AT 250 WORDS)